Lecture-29: Temperature scaling

- untill now, the generated token is selected corresponding to the token longest probability score among all token in the vocabulary.

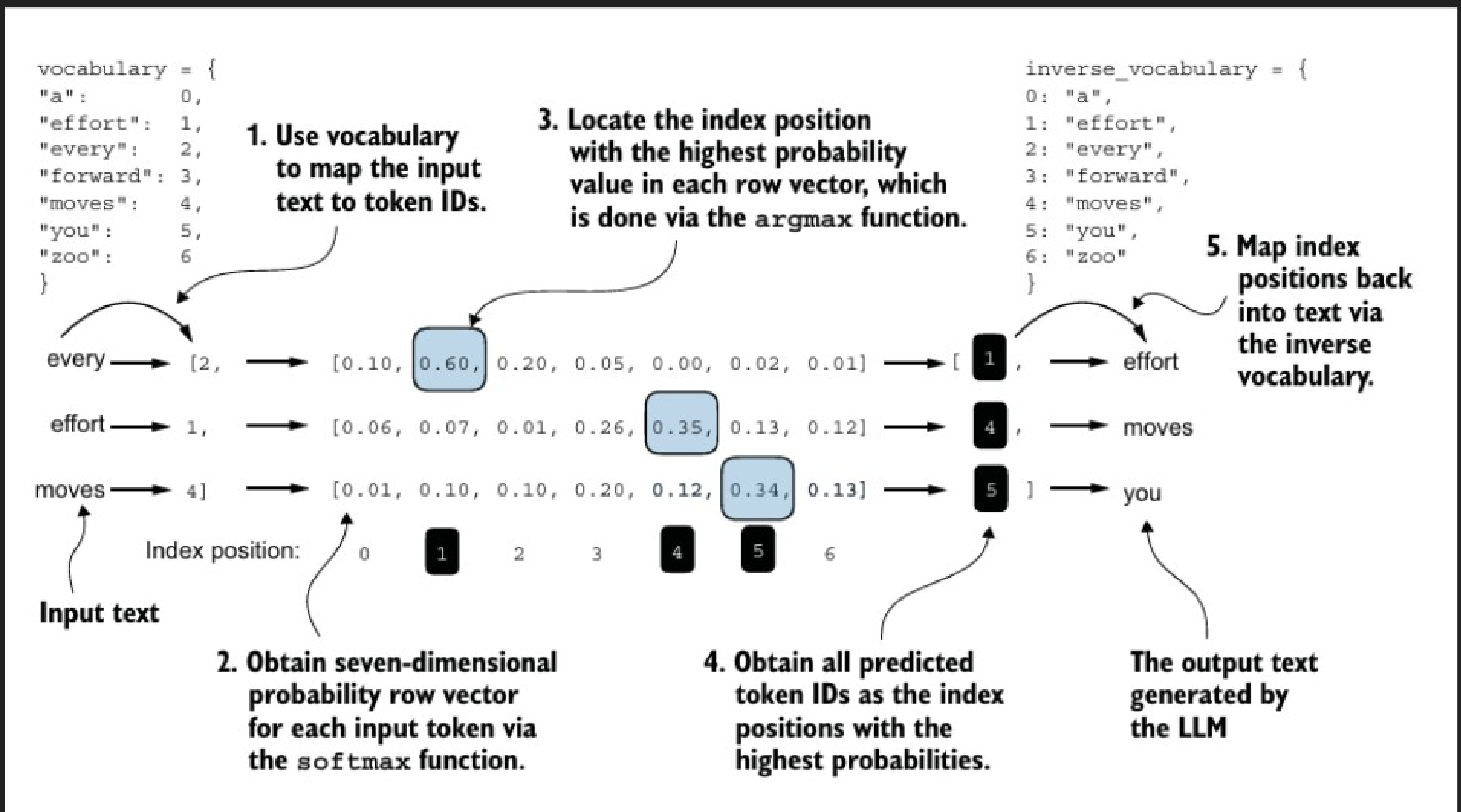


Figure 5.4 For each of the three input tokens, shown on the left, we compute a vector containing probability scores corresponding to each token in the vocabulary. The index position of the highest probability score in each vector represents the most likely next token ID. These token IDs associated with the highest probability scores are selected and mapped back into a text that represents the text generated by the model.

- leads to a lot of randomness and diversity in generated text
- we will learn 2 techniques to control this randmoness

temperature scaling

top-k scaling

Technique 1: Temperature scaling

↳ Replace 'argmax' with probability distribution.

☆ multinomial probability distribution samples next token according to probability score.

* what is temperature: fancy term for dividing the logits by a number greater than zero.

$$\boxed{\text{Scaled logits}} = \frac{\text{logits}}{\text{temperature}}$$

small: sharper distribution.

large: flattened distribution. (more varity, but also more non sense)

Scaled logits → $\boxed{\text{softmax}}$ → Tensor of probability

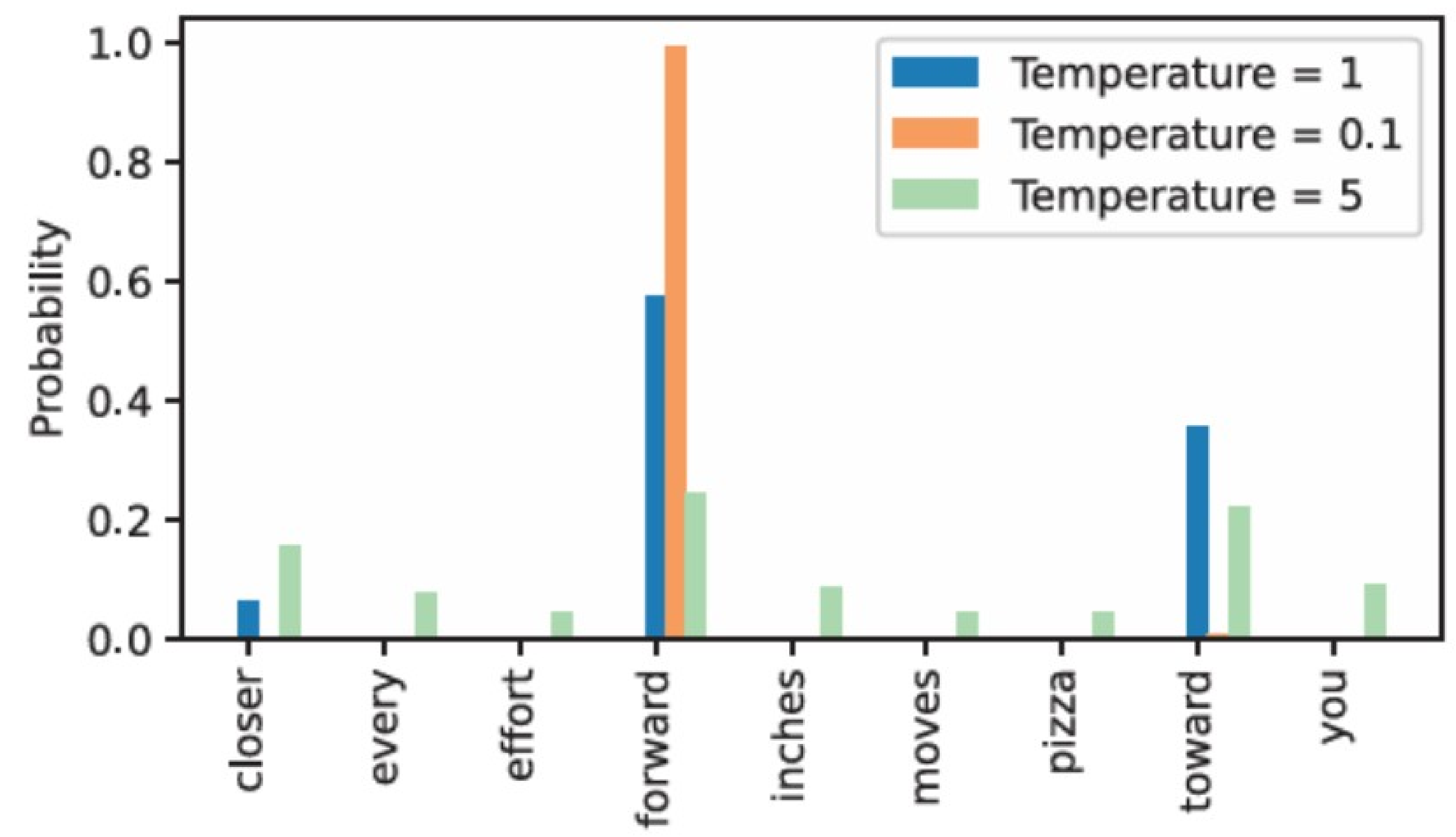


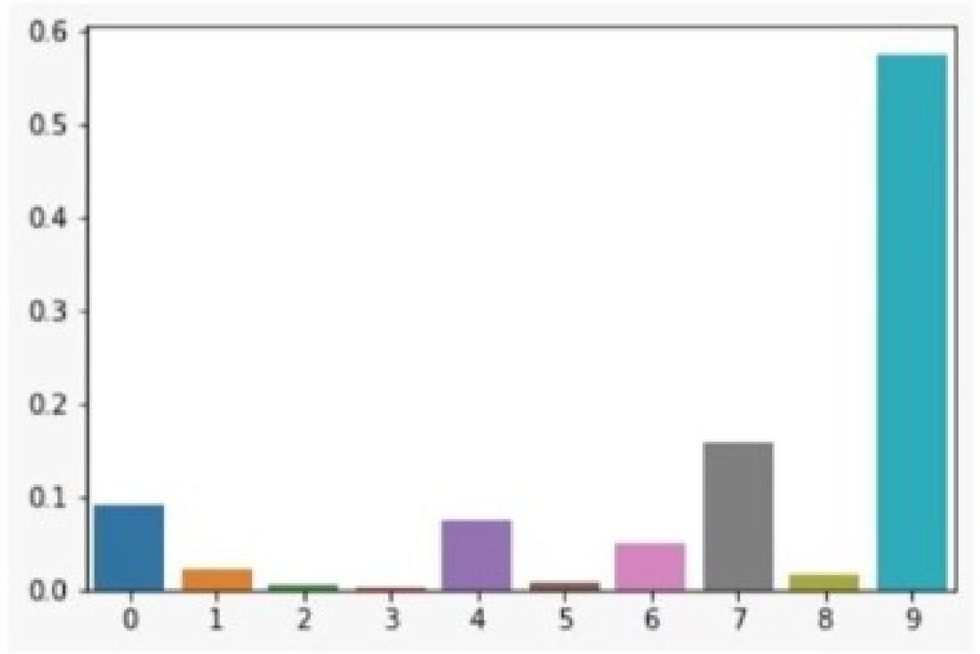
Figure 5.14 A temperature of 1 represents the unscaled probability scores for each token in the vocabulary. Decreasing the temperature to 0.1 sharpens the distribution, so the most likely token (here, "forward") will have an even higher probability score. Likewise, increasing the temperature to 5 makes the distribution more uniform.

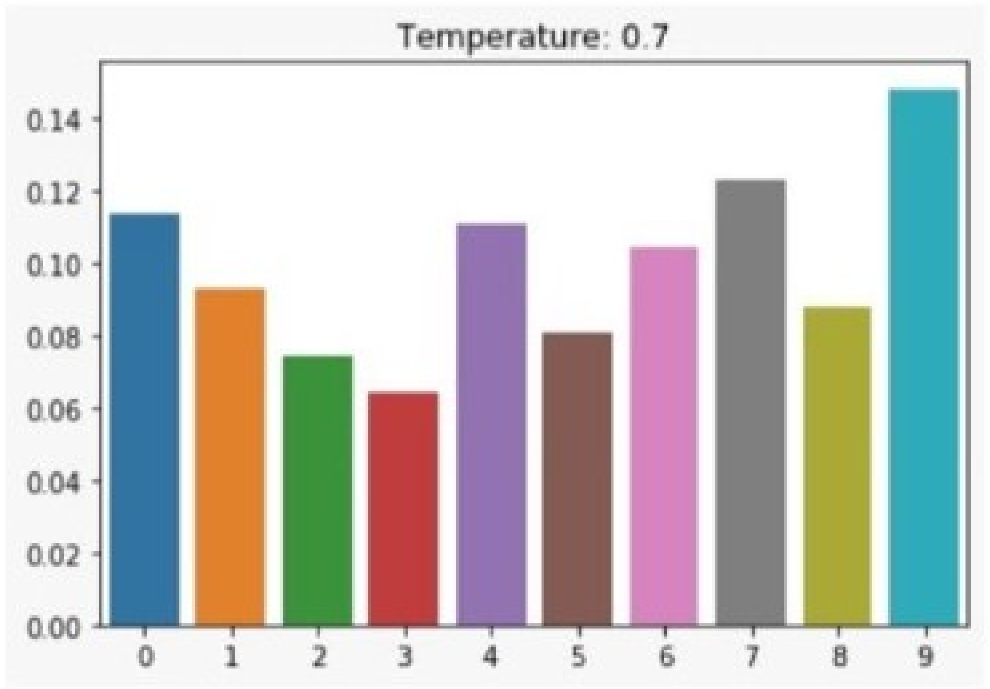
SOFTMAX WITHOUT TEMPERATURE (T=1)

$$\frac{e^{z_i}}{\sum_j e^{z_i}}$$

SOFTMAX WITH TEMPERATURE

$$\frac{e^{z_i/T}}{\sum_j e^{z_i/T}}$$





LESS ENTROPY $\xrightarrow[\text{WITH INCREASE IN T}]{\text{INCREASE IN ENTROPY}}$ MORE ENTROPY